Real-world examples and applications of AI

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2026

Friedman’s theorem

A version of the Friedman’s fundamental theorem of informatics describing the impact of augmented intelligence.

> “A human working in collaboration with an AI-based information system may achieve better performance than the same human working without AI support.”

The theorem is applicable to health care, research, education, and administrative activities.

https://doi.org/10.1197/jamia.M3092

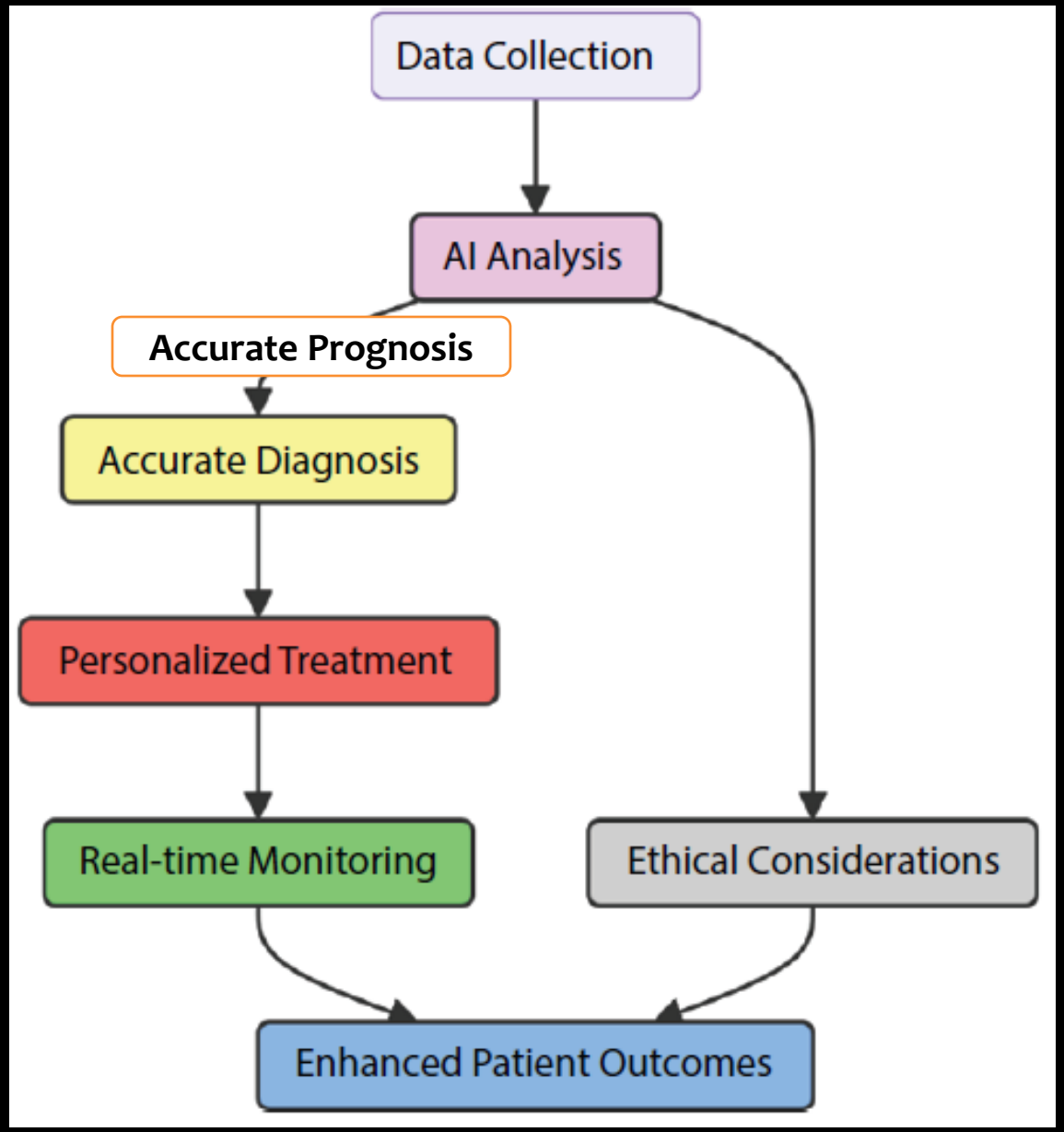
https://doi.org/10.1111/cts.12884

Benefits of AI

- Improved process efficiency: processes become smarter and more efficient, boosting productivity and profits. AI automatises tasks, identifies patterns, optimises workflows and makes real-time adjustments.
- Greater creativity: AI can automatise tedious or repetitive tasks, such as production line work, data gathering, emailing, billing or customer service. This way people can spend more time on more creative or more human-centric tasks.
- Lower risk of error: AI can enter data, do calculations or perform manoeuvres more quickly and more efficiently than humans. In critical sectors, such as healthcare or aviation, AI plays a vital role in assisting with high-risk surgery or predicting technical failures.
- Performance of dangerous tasks: artificial intelligence can assist with tasks that endanger human safety or health, such as rescue operations, bomb disposal, managing contagious or radioactive substances or inspecting hostile environments.
- Resolution of global challenges: AI can tackle complex global challenges such as improving healthcare, addressing climate change and ensuring food security, by for instance enhancing disease prediction, reducing energy use and optimising crop yields.

(https://www.consilium.europa.eu/en/policies/benefits-and-risks-of-ai/#benefits)

AI in Healthcare and Medicine



Modified from : https://doi.org/10.1002/9781394389131.ch17

In medicine, AI is used for prognosis of specific diseases, diagnostic procedures, developing the treatment plan, and enhancing workflow efficiency.

AI in Predictive Analytics (prognosis)

AI can analyze patients' electronic health records (EHRs) to identify individuals at high risk of developing certain diseases, placing them under close medical monitoring — helping doctors detect and treat conditions early, rather than at advanced stages.

According to the Assistant Secretary for Technology Policy (ASTP), **71%** of nonfederal acute care hospitals in the U.S. reported using **predictive AI** integrated into their electronic health records (EHRs) in 2024.

https://www.ncbi.nlm.nih.gov/books/NBK618497/

AI in Medical Imaging Analysis (diagnosis)

AI models can analyse large amounts of information and identify patterns in vast datasets that may be invisible to the human eye.

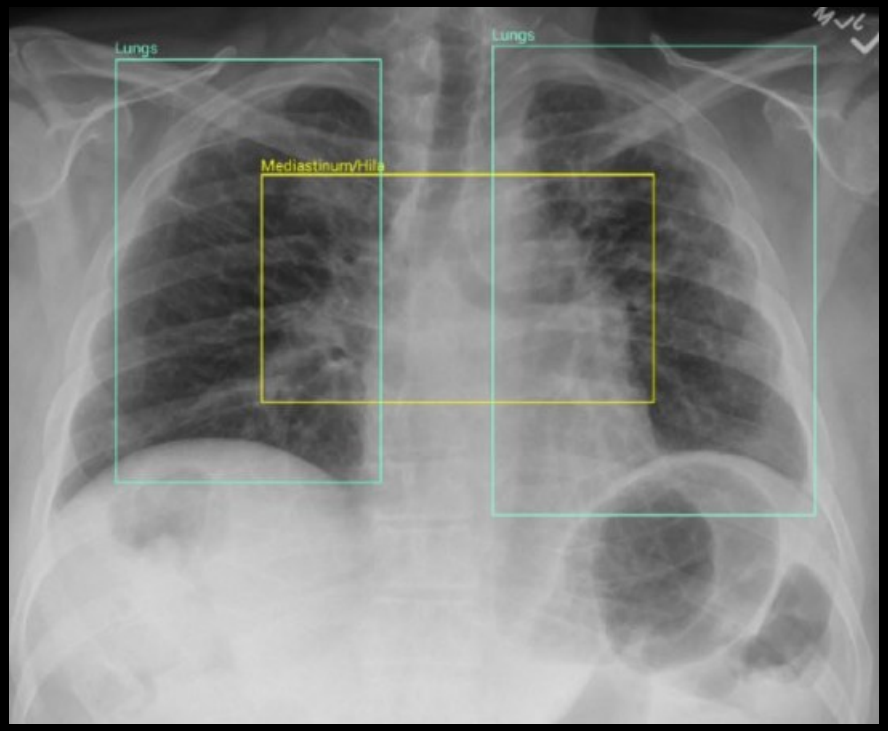
Article | Open access | Published: 24 October 2024

Deep learning improves physician accuracy in the comprehensive detection of abnormalities on chest X-rays

Pamela G. Anderson, Hannah Tarder-Stoll, Mehmet Alpaslan, Nora Keathley, David L. Levin, Srivas Venkatesh, Elliot Bartel, Serge Sicular, Scott Howell, Robert V. Lindsey & Rebecca M. Jones

Scientific Reports **14**, Article number: 25151 (2024) | Cite this article

https://doi.org/10.1038/s41598-024-76608-2



The AI system identified abnormalities that were characterized as regions of interest in Lungs (green boxes) and Mediastinum/Hila (yellow box).

AI in Medical Imaging Analysis (diagnosis)

ELSEVIER

Academic Radiology

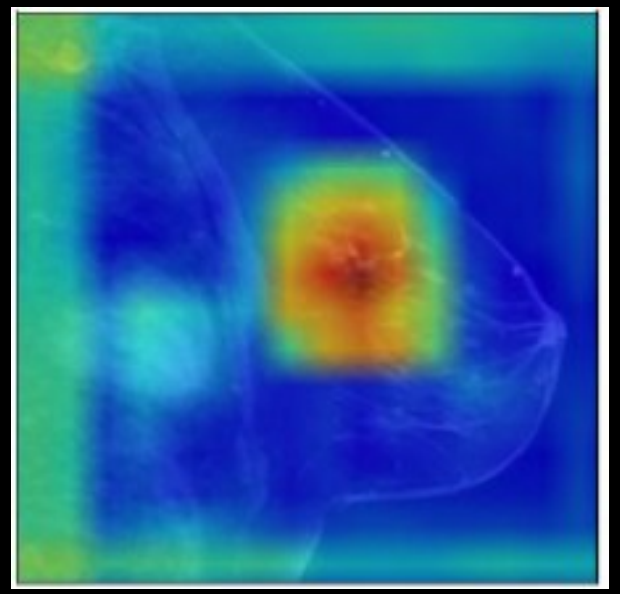
Volume 32, Issue 3, March 2025, Pages 1218-1225

Original Investigation

Early Detection of Breast Cancer in MRI Using AI

Lukas Hirsch [a], Yu Huang [a], Hernan A. Makse [a], Danny F. Martinez [b], Mary Hughes [b], Sarah Eskreis-Winkler [b], Katja Pinker [b], Elizabeth A. Morris [b c], Lucas C. Parra [a], Elizabeth J. Sutton [b]

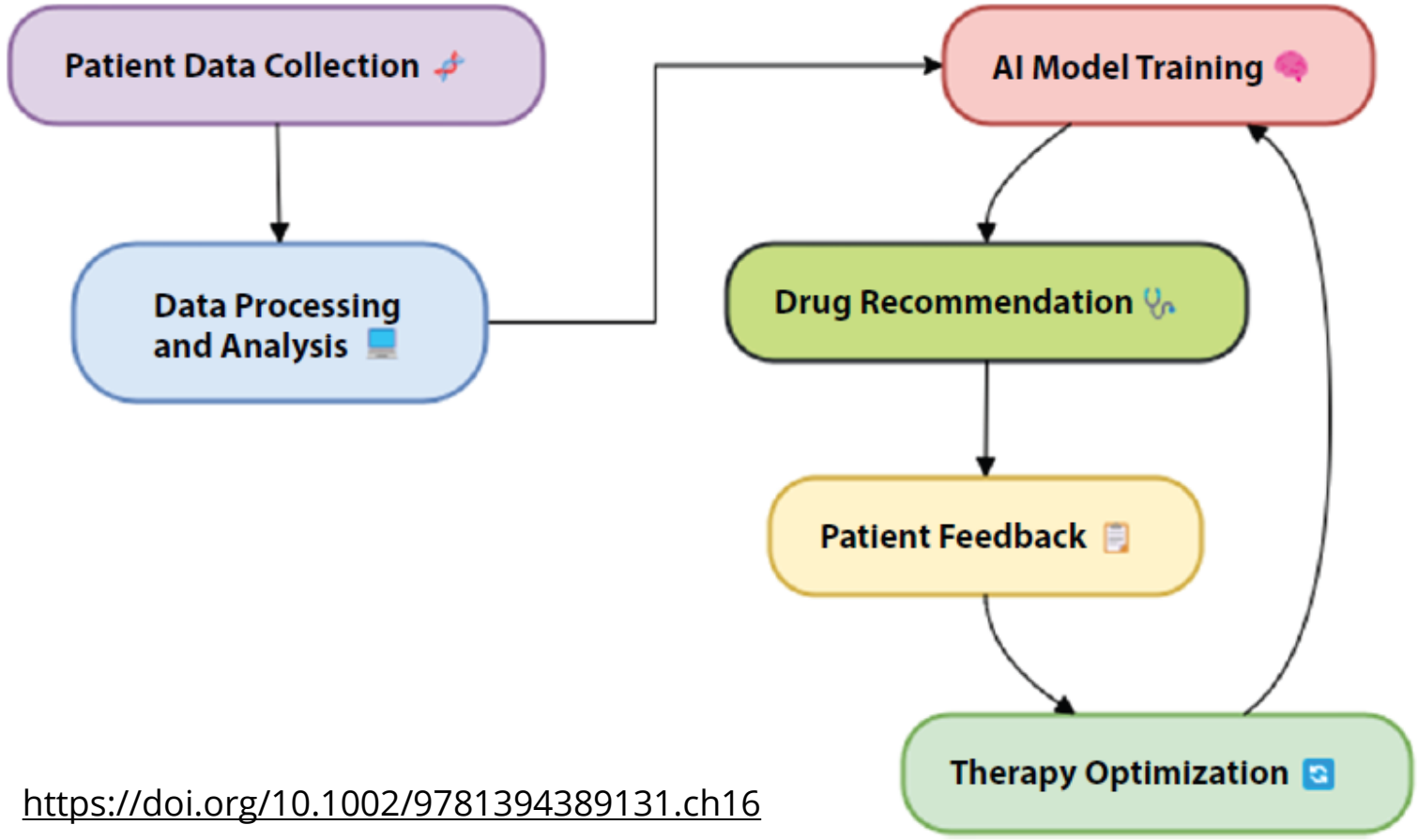
https://doi.org/10.1016/j.acra.2024.10.014



- Retrospective analysis by a radiologist of the top 10% highest risk MRIs as ranked by the AI could have increased early detection by up to 30%.
- AI identified correct anatomic regions in 57% of future cancer cases

Personalized drug therapies using AI

Advanced AI tools have drastically enhanced the effectiveness of treatments tailored to the individual's characteristics, including their genetic makeup, lifestyle, and health conditions.



https://doi.org/10.1002/9781394389131.ch16

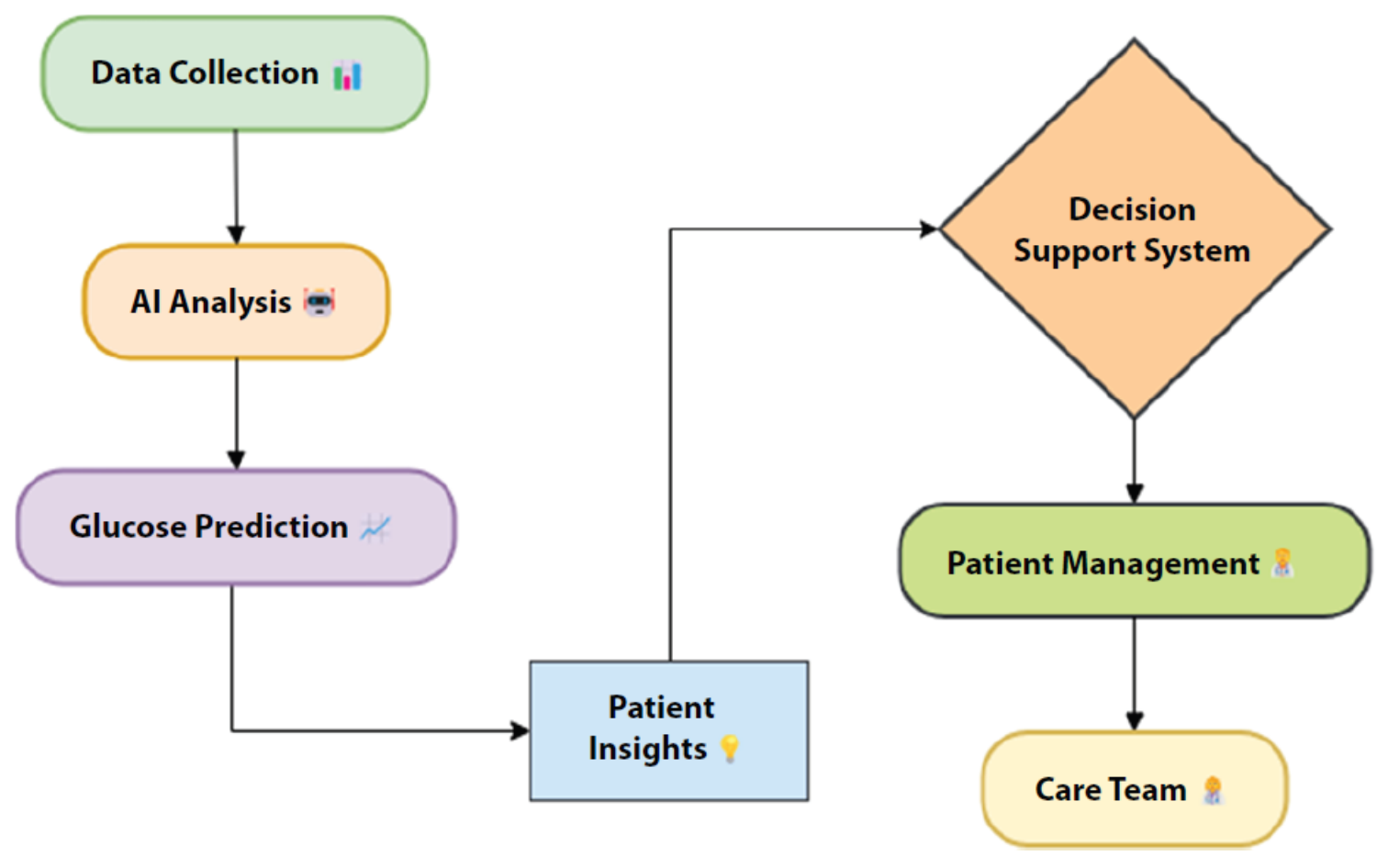
AI's capability of data mining enables identifying the most effective treatments for every individual, thereby improving the efficacy of treatments and minimizing potential adverse reactions.

AI-powered virtual assistants

For example, diabetic patients can receive daily reminders to check their blood sugar levels.

AI-powered virtual assistants have gained popularity, giving users personalized health advices, medication reminders, and even video consultations.

For individuals suffering from chronic illnesses, these virtual health assistants can greatly facilitate the self-management and access to healthcare services.



Robotic-Assisted Surgery

Robots powered by AI are now able to perform surgeries, supporting medical staff and improving patient outcomes.

da Vinci surgical system

https://www.youtube.com/watch?v=N2WW8wLynGE

Versius robotic system

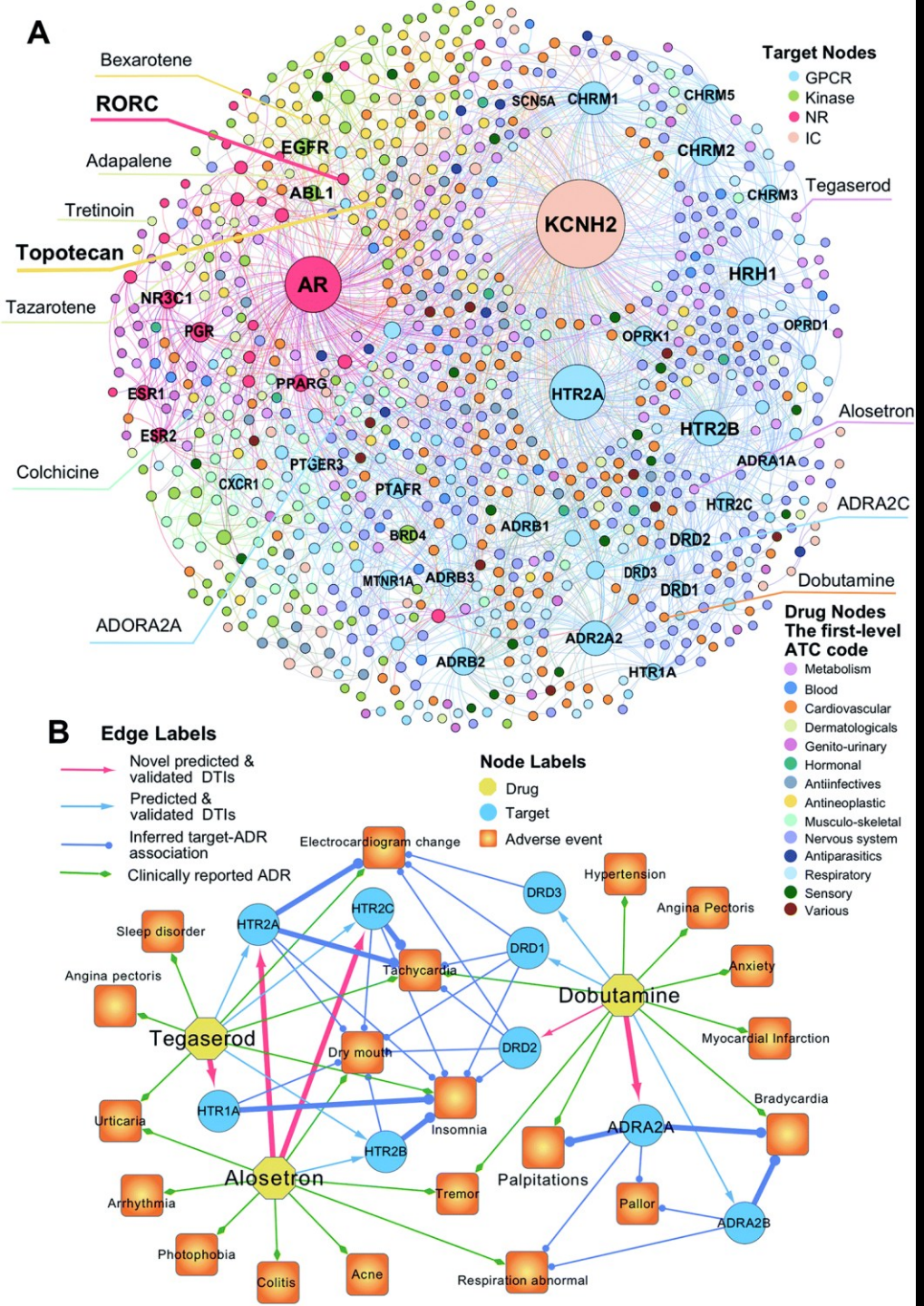
https://www.youtube.com/watch?v=6mhu4E_cZ5Q

Advantages of robotic surgery

- Better visualization as the operating surgeon obtains a three-dimensional image that improves depth perception; camera motion is stable and easily controlled by voice-activated or manual master controls.
- As compared to laparoscopic tools, the manipulation of robotic arm instruments enhances the range of motion, allowing the surgeon to conduct more complicated surgical operations.
- When compared to open and laparoscopic surgery, robotic surgery causes far less physical discomfort. Ergonomic training can aid in the relief and prevention of ergonomic strain during robotic surgery.
- When compared to patients who had open surgery, robotic surgery reduced the chance of readmission by half (52%) and revealed a 77% reduction in the prevalence of blood clots (deep vein thrombosis and pulmonary emboli).

https://doi.org/10.7759/cureus.42370

AI for drug discovery



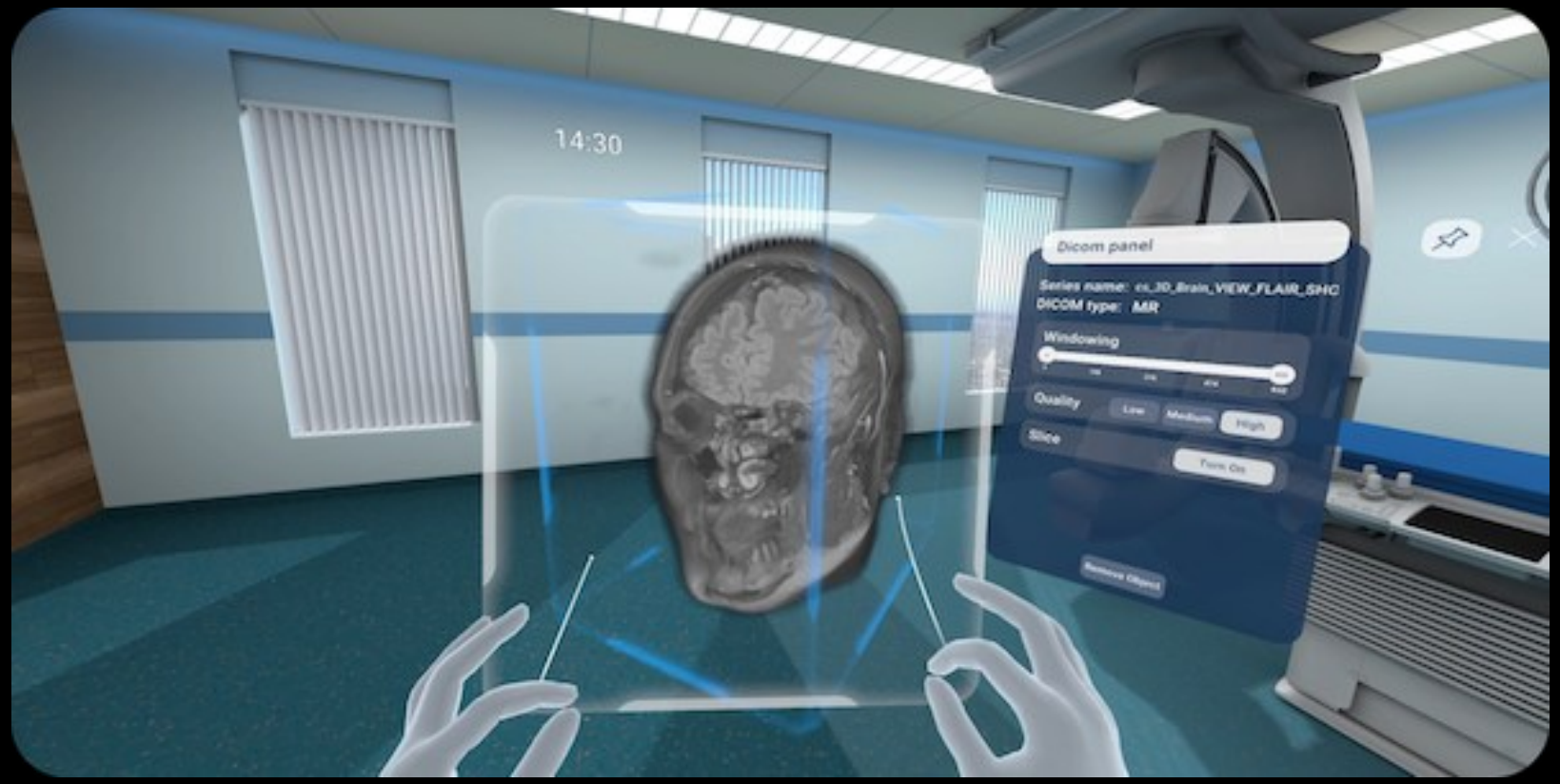
AI algorithms can help:

- Identify new therapeutic uses for existing drugs by analysing large-scale biological and clinical datasets (known as drug repurposing)
- Predict mechanisms of drug toxicity, adverse effects, and potential complications
- Accelerate the design and development of novel drug candidates, thereby reducing research timelines and lowering overall costs in pharmaceutical innovation

(https://pubs.rsc.org/en/content/articlehtml/2020/SC/C9SC04336E)

AI in Medical Education & Training

VR anatomy and surgical labs now use AI to provide real-time haptic feedback, allowing students to practice complex procedures (like endovascular surgery) with "intelligent" resistance and guidance. (https://www.youtube.com/watch?v=fbfLqdhgOW4)



Industrial Automation

AI-driven robots and automation systems are now essential across manufacturing, agriculture, and high-stakes environments. By integrating machine learning with advanced sensors, these systems increase productivity and safety while minimizing human intervention in several keyways:

- **Manufacturing:** Enhancing precision and enabling collaborative workflows between humans and machines (Cobots).
  https://www.youtube.com/watch?v=KIw3iQKC6BE

- **Agriculture:** Automating harvesting and crop monitoring to improve yield and sustainability.
  https://www.youtube.com/watch?v=xzOf4-WaXzE

Humanitarian Aid & Disaster Response

Dangerous Environments: Performing the initial **"high-risk" sweep** of disaster zones—such as firefighting or search-and-rescue—to locate survivors and assess structural integrity before human teams enter.

- **The Taiwan Hualien 7.4 Earthquake (April 2024):** Researchers from the University of Cambridge deployed automated **landslide detection** methods by combining Earth observation (EO) data with AI models. The models identified 7090 landslide events covering > 75 km^2 within ≈ 3 h of the acquisition of the EO imagery.

- **Hurricanes Helene & Milton (2024):** AI used by the State of Florida to process drone and satellite imagery, assessing damage to **21,700 buildings** and 400 miles of roads. The system can map a large neighborhood in **four minutes** using a standard laptop.

https://www.youtube.com/watch?v=qqZJci3C8HQ

https://www.youtube.com/watch?v=0yHjQKXWqYI

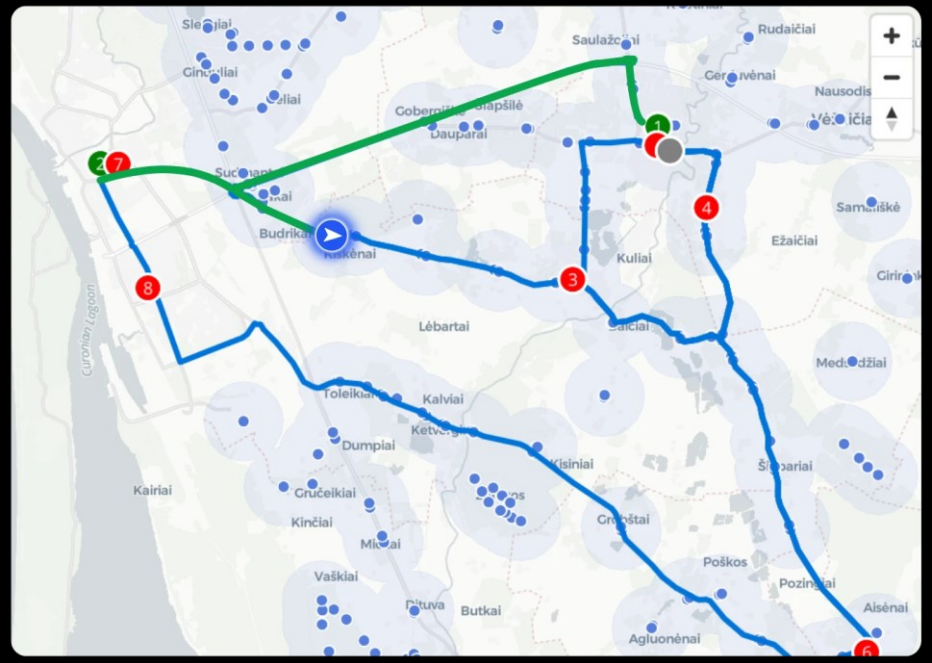
Humanitarian Aid & Disaster Response

https://www.unhcr.org/digitalstrategy/ai-approach/

Transportation and Autonomous Vehicles

Self-Driving Cars: AI powers autonomous vehicles (e.g. robotaxi), with companies such as Tesla or Baidu advancing this technology to overcome barriers and limitations, making transportation safer and more efficient.
(https://www.youtube.com/watch?v=FwABzdr5elM)

Route Planning and Ride-Hailing: AI-powered route planning and ride-hailing are revolutionizing urban mobility by shifting from manual scheduling to automated, real-time decision-making, which enhances efficiency and accessibility, lowers operational costs, environmental impact, and speeds up deliveries or passenger pickups.



The goal is simple: help you arrive at your destination as quickly, safely, and cost-effectively as possible.

An Optimal Route Planner is a tool or software that helps determine the most efficient route between multiple destinations. Whether you need to optimize a single trip or manage the routing for several locations, the software considers various factors like distance, traffic, road conditions, time constraints, and even the type of vehicle to generate the best route possible.

Business, Finance, and Security

- **Financial Services:** AI is used in finance to predict and assess loan risks, automate transactions, and enhance data security by detecting nuanced attacks and improving incident response.

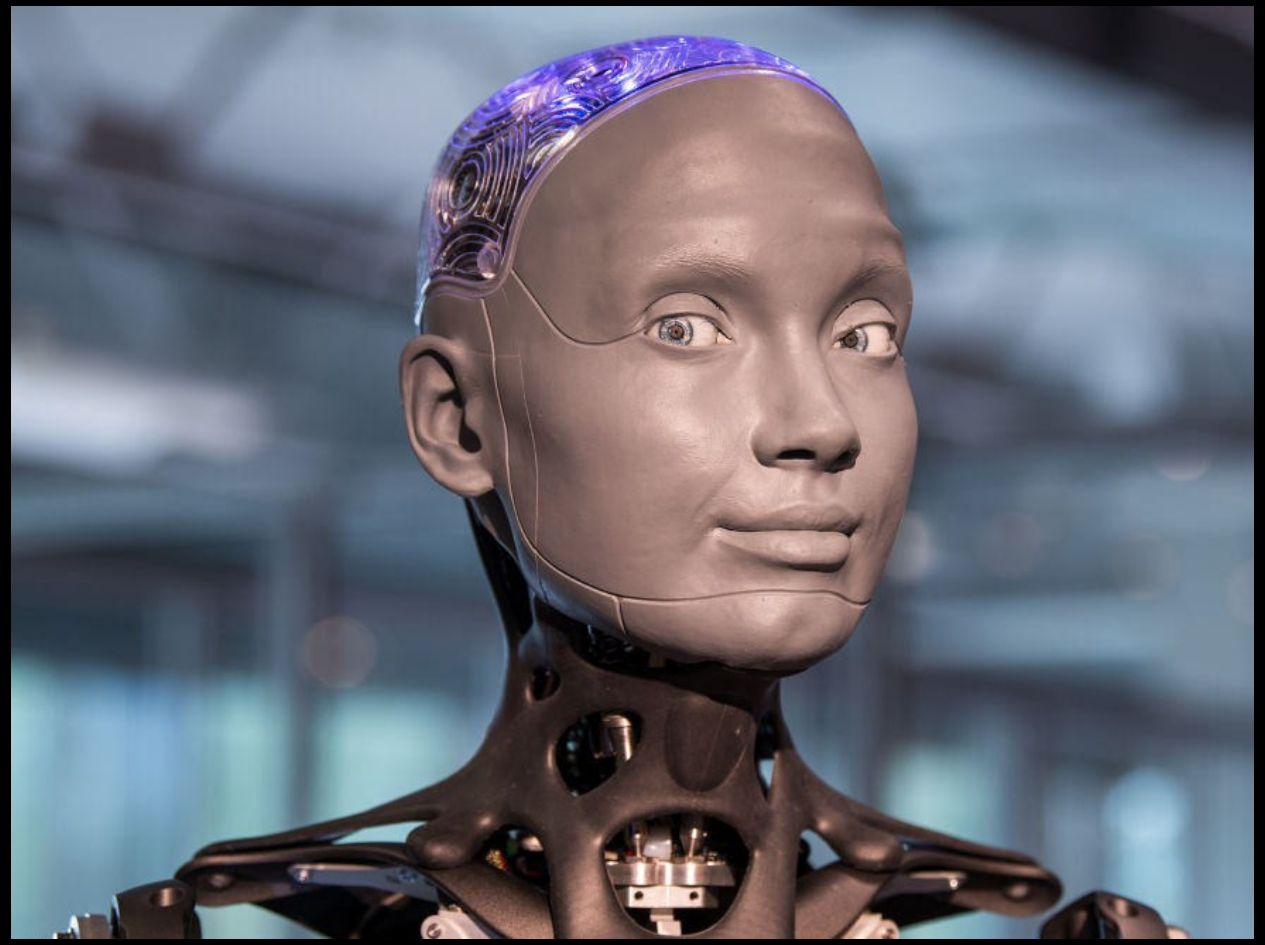
- **E-commerce and Customer Service:** AI-driven recommendation systems, product suggestions, and automated support improve the customer experience in online shopping and other digital services.

- **Cybersecurity:** AI enhances cybersecurity by detecting and flagging malicious activities, such as phishing and fraud, with high accuracy.

Sociable Robots

AMECA

The world's most advanced social humanoid robot.



https://www.youtube.com/watch?v=CGONTgQd8TE

Sociable Robots

Social robots are collaborative robots evolved/specialized in social interaction. They can be used for a variety of purposes; for example, as educational tools and therapeutic aids.

They are capable of:

- Establishing and maintaining social relationships
- Learning and developing social skills, as well as acting as role models
- Using “natural” signals, such as gestures and gaze
- Expressing and perceiving emotions
- Communicating with high-level dialog using natural language
- Expressing a consistent personality and a distinct, recognizable character

https://doi.org/10.3390/healthcare9030244

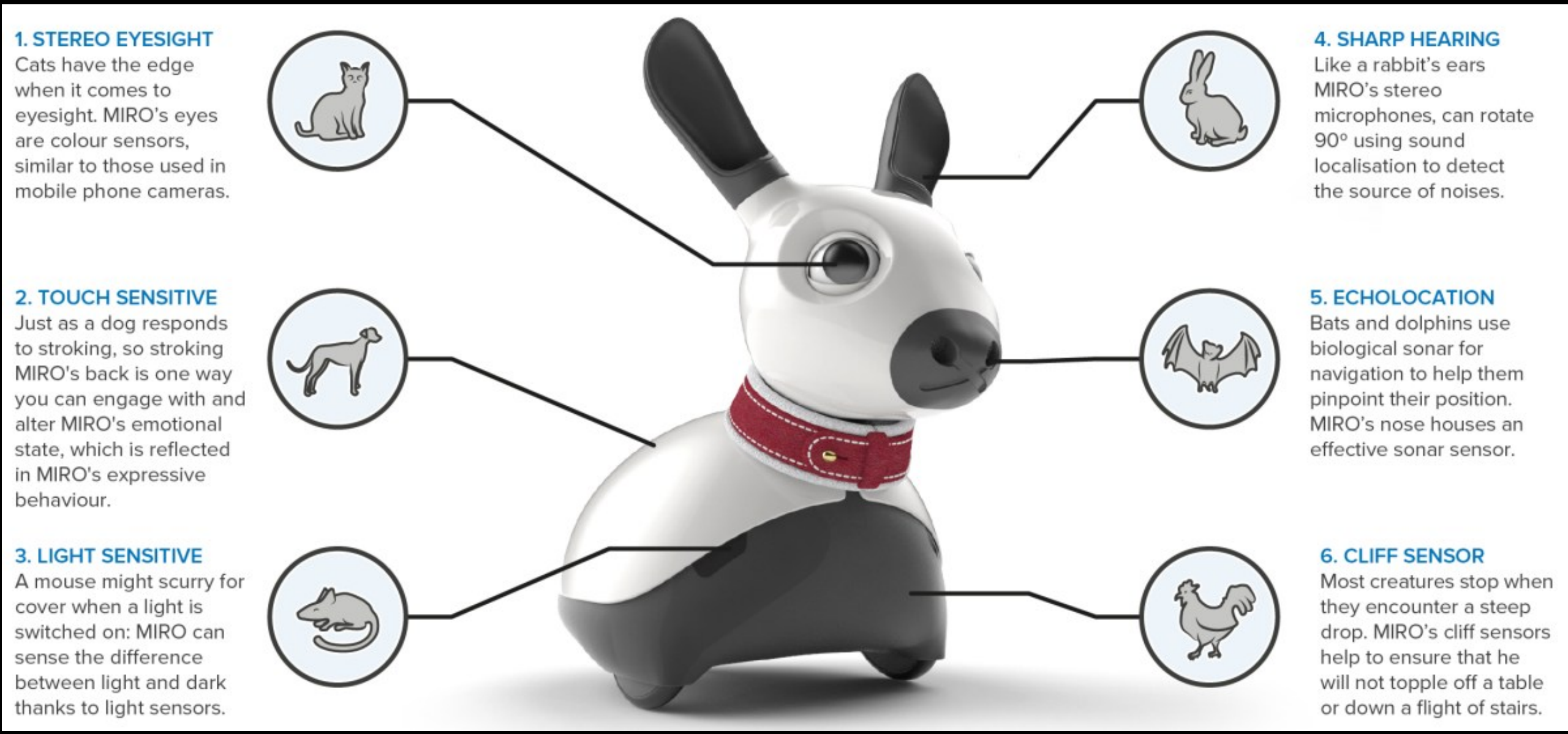
Sociable Robots

Carebots are utilised in hospitals and homes to provide care and support for vulnerable persons such as the elderly and those suffering from physical and mental disabilities. They monitor the health conditions of patients, give medication, manually lift and aid the movements of disabled patients, and provide social companionship. https://doi.org/10.1007/s12369-020-00653-w

Bridging the Loneliness Gap in Elder Care

Friendbots (or companion robot) are companionship-focused social robots that go beyond the typical ‘helper’ roles of carebots; they are specifically designed to support human–robot bonding.

MiRo companion robot



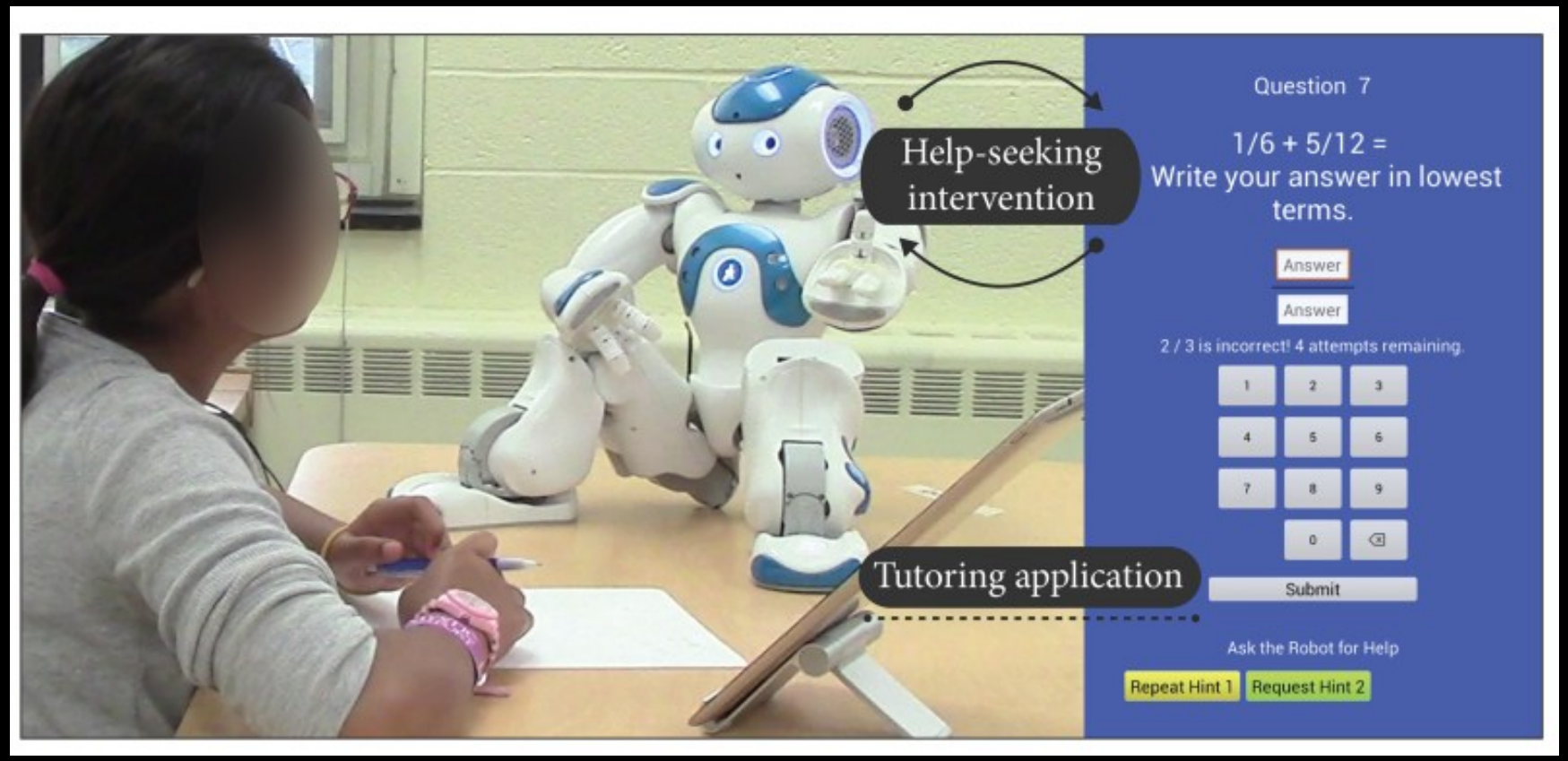
https://consequentialrobotics.com/miro-beta

Robot–Child Tutoring

A Yale-led research team will spend the next five years developing a new breed of sophisticated "socially assistive" robots for helping young children learn to read, appreciate physical fitness, overcome cognitive disabilities and perform physical exercises.

https://www.youtube.com/watch?v=p8m65cawkMc

- **Example of an interactive robot tutoring system to provide help to children practicing math problems.**



https://doi.org/10.1145/3213768

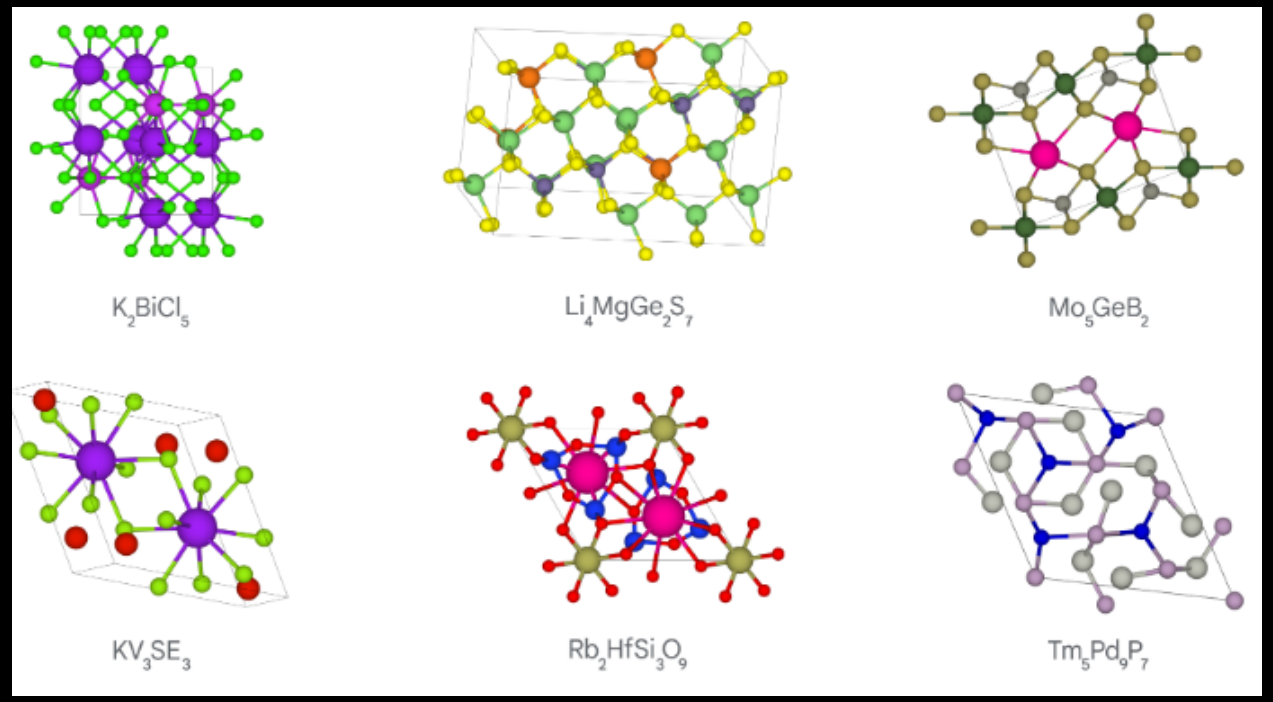
Scientific Discoveries

AI significantly advances scientific knowledge across all fields by accelerating research, generating hypotheses, and enabling insights beyond traditional methods.

Example: Materials discovery

GNoME is one of the most remarkable demonstrations of AI accelerating science. GNoME stands for Graph Networks for Materials Exploration — a deep learning tool developed by Google DeepMind that dramatically increases the speed and efficiency of materials discovery by predicting the stability of new materials.

External researchers in labs around the world have independently created 736 of these new structures experimentally in concurrent work.



https://www.nature.com/articles/s41586-023-06735-9

AI Won Gold at the Mathematical Olympiad

Artificial intelligence has demonstrated revolutionary progress in mathematical reasoning.

Multiple different neural reasoning models – Google DeepMind's Gemini and OpenAI's undisclosed GPT model – announced gold medal performance on the 2025 International Math Olympiad (IMO).

Winning Gold at IMO 2025 with a Model-Agnostic Verification-and-Refinement Pipeline*

Yichen Huang (黄溢辰)† Lin F. Yang (杨林)‡

October 1, 2025

Abstract

The International Mathematical Olympiad (IMO) is widely regarded as the world championship of high-school mathematics. IMO problems are renowned for their difficulty and novelty, demanding deep insight, creativity, and rigor. Although large language models perform well on many mathematical benchmarks, they often struggle with Olympiad-level problems. Using carefully designed prompts, we construct a model-agnostic, verification-and-refinement pipeline. We demonstrate its effectiveness on the recent IMO 2025, avoiding data contamination for models released before the competition. Equipped with any of the three leading models—Gemini 2.5 Pro, Grok-4, or GPT-5—our pipeline correctly solved 5 out of the 6 problems ($\approx$ 85.7% accuracy). This is in sharp contrast to their baseline accuracies: 31.6% (Gemini 2.5 Pro), 21.4% (Grok-4), and 38.1% (GPT-5), obtained by selecting the best of 32 candidate solutions. The substantial improvement underscores that the path to advanced AI reasoning requires not only developing more powerful base models but also designing effective methodologies to harness their full potential for complex tasks.

https://arxiv.org/pdf/2507.158

Solving Problems in Mathematics

FrontierMath includes both carefully crafted challenge problems and open research problems that remain unsolved by mathematicians.

FRONTIERMATH: A BENCHMARK FOR EVALUATING ADVANCED MATHEMATICAL REASONING IN AI

Elliot Glazer[1†‡], Ege Erdil[1†], Tamay Besiroglu[1†], Diego Chicharro[2‡], Evan Chen[3‡], Alex Gunning[4‡], Caroline Falkman Olsson[1], Jean-Stanislas Denain[1], Anson Ho[1], Emily de Oliveira Santos[5‡], Olli Järviniemi[†], Matthew Barnett[1], Robert Sandler[1], Matej Vrzala[1], Jaime Sevilla[1], Qiuyu Ren[6‡], Elizabeth Pratt[6‡], Lionel Levine[7‡], Grant Barkley[8‡], Natalie Stewart[8‡], Bogdan Grechuk[9‡], Tetiana Grechuk[9‡], Shreepranav Varma Enugandla[6‡], Mark Wildon[10‡], Terence Coelho[15‡], Ahsan Z. Khan[8‡], Will Brian[11‡], Pedro Teixeira[12‡], Vinh-Kha Le[6‡], Jan Jurka[13‡], Johannes Schmitt[16‡], Sai Sanjeev Balakrishnan[6‡], Peisheng Yu[6‡], David Brodsky[14‡], Razzi Masroor[3‡], Jessica Wang[17‡] Daniel Arreola[18‡], Joseph Knight[14‡], Noah Lebowitz-Lockard[14‡], Tasos Moulinos[19‡], Josh Ducey[20‡]

https://arxiv.org/pdf/2411.04872

The most difficult math benchmark to date, FrontierMath, created by experts to maximally challenge modern AI systems at mathematical reasoning, has seen many of its highest tier problems solved by AI.

Solving Problems in Mathematics

First-Proof, a set of 10 novel research-grade problems that were held-out intermediate results in the work of expert mathematicians saw the majority of its problems solved autonomously within a week of release.

Google DeepMind 2026-3-2

Aletheia tackles *FirstProof* autonomously

Tony Feng*, Junehyuk Jung, Sang-hyun Kim, Carlo Pagano, Sergei Gukov, Chiang-Chiang Tsai, David P. Woodruff, Adel Javanmard, Aryan Mokhtari, Dawsen Hwang, Yuri Chervonyi, Jonathan N. Lee, Garrett Bingham, Trieu H. Trinh, Vahab Mirrokni, Quoc V. Le, Thang Luong*

*Project leads. Work conducted under Google DeepMind.

We report the performance of *Aletheia* (Feng et al., 2026b), a mathematics research agent powered by Gemini 3 Deep Think, on the inaugural *FirstProof* challenge. Within the allowed timeframe of the challenge, *Aletheia* autonomously solved 6 problems (2, 5, 7, 8, 9, 10) out of 10 according to majority expert assessments; we note that experts were not unanimous on Problem 8 (only). For full transparency, we explain our interpretation of *FirstProof* and disclose details about our experiments as well as our evaluation. Raw prompts and outputs are available at https://github.com/google-deepmind/superhuman/tree/main/aletheia.

https://arxiv.org/pdf/2602.21201

Solving Problems in Theoretical Physics

Researchers in mathematics, computer science, and theoretical physics have found modern AI systems useful in practice for developing proofs and offering high-level reasoning to assist their work.

Solving an Open Problem in Theoretical Physics using AI-Assisted Discovery

Michael P. Brenner[1,2], Vincent Cohen-Addad[1], and David P. Woodruff[1,3]

[1]Google Research
[2]School of Engineering and Applied Sciences, Harvard University
[3]School of Computer Science, Carnegie Mellon University and Google Research

March 6, 2026

https://arxiv.org/pdf/2603.04735